

Exercise 2 - Potential energy surface of the H-F molecule, four computational methods

a) Potential energy curves

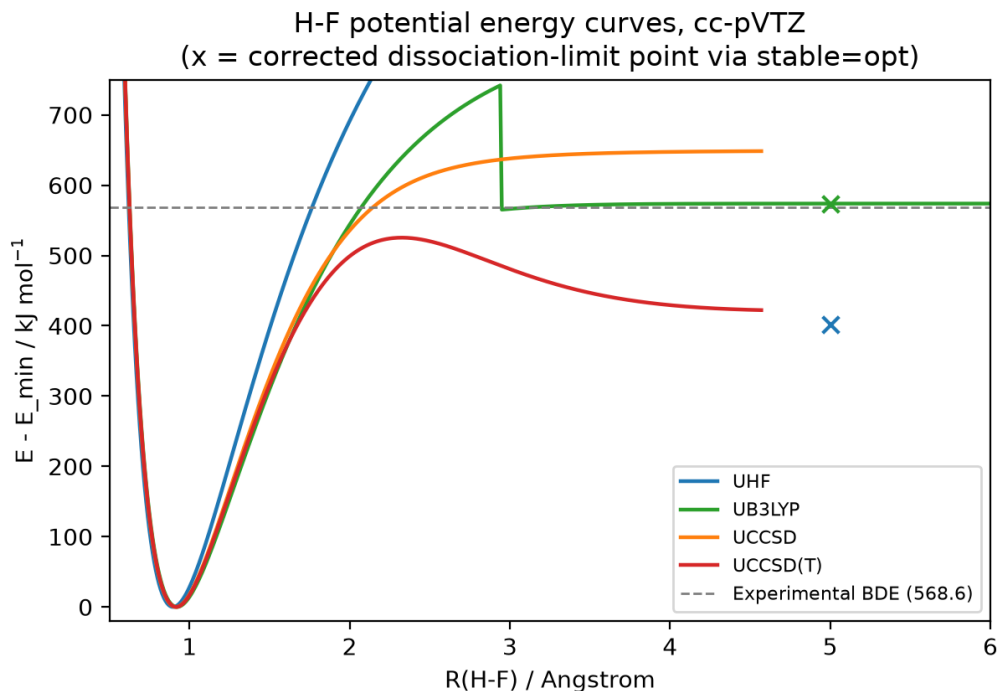


Figure 1. H-F potential energy curves, UHF/UB3LYP/UCCSD/UCCSD(T), cc-pVTZ.

b) Equilibrium bond length R_e

Method	$R_e / \text{Angstrom}$
UHF	0.90
UB3LYP	0.92
UCCSD	0.92
UCCSD(T)	0.92

c) Dissociation energy D_e

Method	$D_e / \text{kJ mol}^{-1}$
UHF	402.0
UB3LYP	573.8
UCCSD	648.4
UCCSD(T)	422.3

UCCSD and UCCSD(T) values are approximate: the amplitude equations failed to converge beyond $R = 4.57 \text{ \AA}$, so the last converged point was used in place of the true dissociation limit.

d) Comparison with the experimental bond dissociation energy (568.6 kJ/mol)

Method	D_e / kJ mol ⁻¹	Difference from experiment
UHF	402.0	-166.6
UB3LYP	573.8	+5.2
UCCSD	648.4	+79.8
UCCSD(T)	422.3	-146.3

UB3LYP is closest to experiment (off by only 5.2 kJ/mol) and is therefore the most accurate of the four methods here; UHF and UCCSD(T) underestimate D_e by over 140 kJ/mol and UCCSD overestimates it by 79.8 kJ/mol.